Kaggle Project (binome)

Deep learning models are widely used for image classification. The goal of this project is to compare state-of-the-art models on the TILDA textile texture dataset. The link of the project is : `https://www.kaggle.com/t/286b14f08b6941f3a1bb4b25e23f6f78`.

Notice : You should upload a report in **English**, together with your code to Moodle, for the validation of the course. Do not copy-paste other people's report or code.

Deadlines : Kaggle submission : 25 June 2026 | Report : 26 June 2026.

1 Learn how to use Pytorch

Refer to the website `https://pytorch.org/tutorials/beginner/basics/intro.html`.

2 Train state-of-the-art CNN models

The goal is to achieve a good classification accuracy on the test dataset, using Pytorch.

Question 2.1 : Pre-process your images so that they have the same input size to your model, e.g. use data augmentation.

Question 2.2 : Implement one or two models from the following list :
- LeNet Model [1]
- AlexNet Model [2]
- ResNet Model [3]
- or any choice available in the literature.

In your report, you should explain
- Why did you choose this particular network ?
- Give a precise definition of the model that you use, e.g. the number of layers, the type of each layer in the CNN, and a brief description on what is each layer doing.

Question 2.3 : Train your model using mini-batch SGD. Specify the optimization method which you use and report the total training time. To reduce the training time, you may use a GPU card.

Question 2.4 : Perform your parameter turning on a validation set to avoid over-fitting. Summarize your results in table/figure.

3 Machine learning and AI safety

3.1 Bias introduction in TILDA

We try to understand how an image classification model (such as the ones trained in section 2) can be biased towards a specific outcome. To make things easier we consider the binary classification problem (2 classes : cauliflower and head cabbage).

Question 3.1 : Describe, with your own words, a real life situation in which there is, or there might be, bias. Comment on the consequences.

We propose to implement the following setup :

3.1.1 Biased Dataset Construction

We decide to concatenate to each of our images x from TILDA a new set of variables ϵ. As we shall specify, the second variable ϵ is a noise-like image, which is not directly computed from $x \in \mathbb{R}^{N \times N}$. Yet, it can introduce some bias if we choose the value of $\epsilon \in \mathbb{R}^{N \times N}$ to be strongly correlated to the label $y \in \{0, 1\}$ of each image x.

More precisely, we assume $(X, y) = ([x, \epsilon], y) \in \mathbb{R}^{2 \times N \times N} \times \{0, 1\}$ is a random sample of the modified dataset. We shall specify the value of ϵ using y. Let $p_0 \in [0, 1]$, $p_1 \in [0, 1]$ be two probabilities. Given (x, y), the bias variable S is defined as

$$S|\{y = k\} \sim Bernoulli(p_k), \quad k \in \{0, 1\}.$$

This means that $\mathbb{P}(S = 1|y = k) = p_k$ and $\mathbb{P}(S = 0|y = k) = 1 - p_k$. Then we choose ϵ according to S as below :

— $\epsilon = 0$ if $S = 0$

— $\epsilon \sim \mathcal{N}(0, I)$ if $S = 1$

where I is the identity matrix of size $N \times N$.

In the extreme case where $p_0 = 0$ and $p_1 = 1$, one can ignore the original image x and only use ϵ to predict y. In that case, our predictions are never using the image x and we are only relying on the noise-like ϵ that we introduced to the dataset.

Advice : in Pytorch, we can build a biased dataset by simply adding the biased variables ϵ as a dedicated channel to each original image x in the whole dataset.

Question 3.2 : Given p_0 and p_1, build a biased dataset from TILDA using the approach described in 3.1.1.

3.1.2 Model Bias Evaluation

We compare two different settings :

— $model_1$ is trained on an unbiased TILDA dataset ($p_0 = 1/2, p_1 = 1/2$)

— $model_2$ is trained on a biased version of TILDA ($p_0 = 0, p_1 = 1$)

We finally compare the actual bias in both $model_1$ and $model_2$.

Assume $\widehat{y}(X)$ is the prediction of a model computed from X. We shall separate the dataset set into 2 groups, one with $S = 0$, the other with $S = 1$. The bias of this model can be computed from a ratio of these two groups, using the DI metric [4] (at the following link : `https://github.com/wikistat/Fair-ML-4-Ethical-AI/tree/master/Propublica`) defined as :

$$DI = \frac{\mathbb{P}(\widehat{y}(X) = 1|S = 0)}{\mathbb{P}(\widehat{y}(X) = 1|S = 1)}$$

In practice, a model is considered unbiased as long as the DI metric is close to 1.

Question 3.3 : Study experimentally if $model_2$ is biased or not on test data. Compare you results to what you have with $model_1$.

Question 3.4 : Compare the accuracy scores on your test data for both $model_1$ and $model_2$ for the binary classification task.

Question 3.5 : Summarize your results into tables (e.g. $model_1$ and $model_2$, vs. DI and accuracy evaluated on training/validation sets). What can you conclude from the results ?

3.2 Bias Study in the Literature

Question 3.6 (bonus) : This article gives an overview of current AI system requirement in EU [4]. Read it carefully and write a short essay about one page through your critical thinking on “Est-ce qu’on peut faire confiance à mon modèle ?” from the perspective of model bias. You can also use other articles in the literature. More broadly, you may refer to the reports of CNIL [5].

Références

[1] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11) :2278–2324, 1998.

[2] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6) :84–90, 2017.

[3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

[4] `https://hal.science/hal-03253111`.

[5] `https://www.cnil.fr/fr/intelligence-artificielle/guide/conformite-des-systemes-dia-les-autres-guides-outils-et-bonnes-pratiques`.